Recursive Multi-Teacher Distillation for 2v2 Soccer

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Abstract: We train cooperative agents for Unity ML-Agents Soccer-Twos (2v2) using PPO on Ray RLlib. Two modifications are applied: a v2 potential-based reward-shaping recipe with five dense signals on top of the sparse goal reward, and a team-level Siamese + cross-attention policy trained via multi-stage knowledge distillation. Our agent is produced by a 3-stage pipeline: (i) distill a 3-member deploy-time ensemble into an 031B-architecture student, (ii) refresh the teacher pool recursively with the Gen-1 output plus an LR-diverse variant, (iii) extend training past teachers' convergence horizons. Our agent scores 10/10 matches vs. Baseline, 10/10 vs. Random, and 10/10 vs. the TA competitive-agent bonus (goal ratios 9.4 : 1, 215.5 : 1, and 20.9 : 1; Table 4), clearing every rubric threshold. Source code, experiment snapshots, and evaluation logs are available at https://github.com/Redemption-ZTX/cs8803.

Keywords: Multi-agent RL, PPO, Knowledge Distillation, Reward Shaping, Soccer-Twos

1 Method

Environment and algorithm. Soccer-Twos [1] has 336-dim ray-cast observations per agent and a factored MultiDiscrete([3, 3, 3]) action (flattened to Discrete(27)). We train with single-team PPO [2] on Ray RLlib 1.4 [3] with GAE [4]. The team objective adds a decaying KL distillation term in the style of Born-Again Networks [5]:

$$\mathcal{L}(\theta) = \mathbb{E}_t\big[\min\big(\rho_t A_t,\ \mathrm{clip}(\rho_t, 1\pm\epsilon)A_t\big)\big] + c_v\mathcal{L}^{VF} + \alpha(t)\,\mathrm{KL}(\pi_\theta \,\|\, \bar{\pi}_{\mathrm{teach}})\,, \tag{1}$$

where $\bar{\pi}_{\mathrm{teach}}$ is the factor-wise probability average of frozen teachers and $\alpha(t) : 0.05 \to 0$ decays linearly over 8000 updates.

Architecture (Figure 1). The policy is a Siamese network [6]: two shared-weight per-agent encoders ($336 \to 256 \to 256$) process each teammate's raycast independently, followed by a cross-agent attention block [7] with 4 learned tokens of dimension 64 that lets either teammate condition on the other's features before the merge layer ($256\times2 \to 256 \to 128$) and the joint-action head. A shared value head re-uses the merge features ($\sim$0.46M total parameters).

Project lineage (Table 1). The final pipeline did not arrive as one lucky jump — it is the last three rows of an eight-checkpoint chain in which each step removed one specific bottleneck. Per-agent PPO with v2 shaping produced playable but sub-0.87 policies; the team-level re-formulation (shared teammate encoders + 4-token cross-attention, the "031B" block in Figure 1) broke the per-agent ceiling at 0.882; Stages 1–3 of Figure 2 are the final three rows of Table 1, converting teacher diversity into a single deployable policy.

Reward modification — v2 shaping. Five potential-based terms (Table 2) augment the sparse $\{\pm1\}$ goal reward; all scales are $O(10^{-3})$ so shaping cannot dominate. We ablated event rewards (shot/tackle/clearance) and goal-proximity and found them harmful in this environment (full matrix in [8]).

Deploy-time ensemble 034E (Stage-1 distillation target). Panel (a) of Figure 2 summarises the three teachers. `031B@1220` is a scratch 031B model with v2 shaping. `045A@180` adds a

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Table 1: Main milestones before the final recursive-distill pipeline. Each row removed one bottleneck rather than adding a new trick. The last three rows correspond exactly to Stages 1–3 of Figure 2.

Stage	Checkpoint	WR	Main contribution
BC + PPO	`017@2100`	0.842	first reliable learning signal
Stable roles	`025b@80`	0.842	fewer unstable PPO updates
Per-agent ceiling	`029B@190`	0.868	best local per-agent optimum
Team encoder	`031A@1040`	$\sim$0.860	teammate-symmetric prior
Cross-attention	`031B@1220`	0.882	best scratch single model
Distillation (S1)	`055@1150`	0.907	main single-policy lift (+0.027)
Recursive distill (S2)	`055v2@1000`	0.909	SOTA-tier plateau
Late extend (S3)	`055v2_extend@1750`	**0.907**	most validated submission

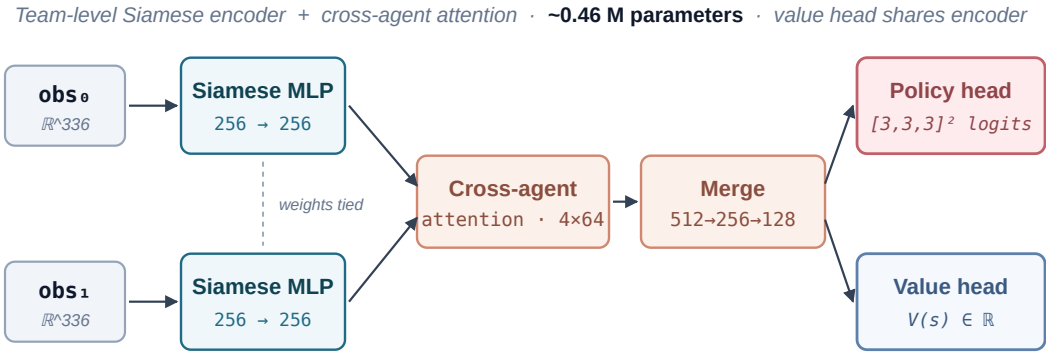

Figure 1: Team-level Siamese + cross-attention policy ("031B" architecture, $\approx$ 0.46M params). Shared-weight per-agent encoders are invariant to teammate index; a 4-token cross-attention block fuses the two streams into the merge, which feeds a joint-action policy head and a value head.

learned W/L reward head — a small MLP $f_\phi(s) \in [0,1]$ trained by binary cross-entropy on 184k per-state labels harvested from earlier 031-series self-play episodes (each state inherits its episode's $\{0,1\}$ win label). `051A@130` adds a PBRS potential $\Phi_\psi(s)$ [9]: we captured $\sim 10^3$ strong-vs-strong episodes (031B vs. 031A) and regressed a small MLP to the final outcome of each state, then added the policy-invariant shaping $r'(s,s') = \gamma\Phi_\psi(s') - \Phi_\psi(s)$ on top of v2. Both learned-reward predictors trained their respective teacher individually and are not re-used for 055 / 055v2 / 055v2_extend — their signal enters the 055-series only through the teacher-ensemble soft targets. The factor-averaged 034E ensemble itself scores only 0.878 — below its best constituent, because the three members share the same raycast-encoder family and their errors correlate. But their per-episode failure modes are orthogonal (defensive pin vs. wasted possession vs. progress deficit), which is the lift our Stage-1 student captures: a single network can condition its reaction on observation; a fixed average cannot.

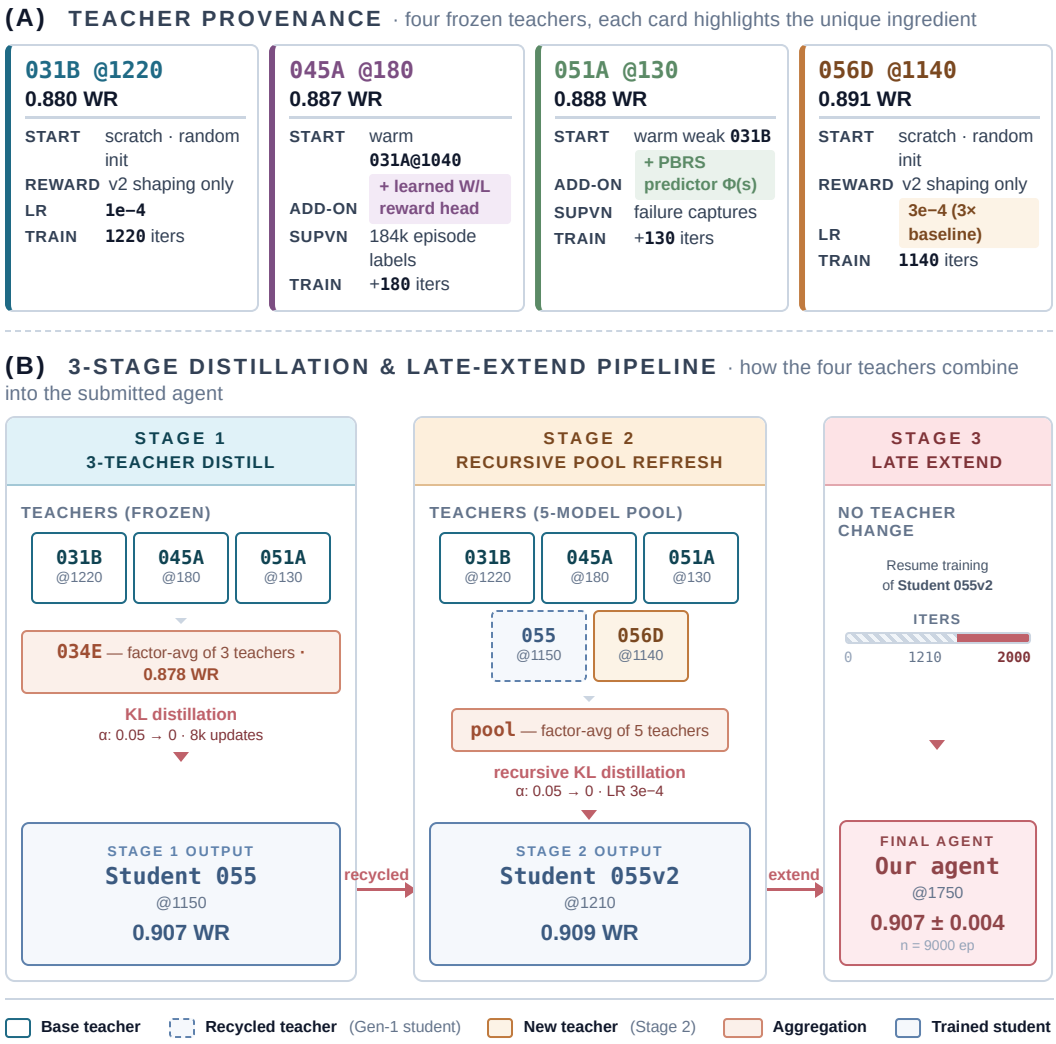

Figure 2: Three-stage training pipeline. Stage 1 distills the 3-member 034E ensemble (0.878 WR) into a student matching 0.907; Stage 2 refreshes the teacher pool with the Gen-1 student and an LR-diverse variant; Stage 3 extends past the original iter-1250 horizon to iter 2000, reducing variance.

Table 2: v2 reward shaping (per-step).

Term	Value
time_penalty	0.001
ball_progress	0.01
opp_progress_penalty	0.01
possession_bonus ($\leq$ 1.25u)	0.002
deep_zone_outer ($x \leq -8$)	-0.003
deep_zone_inner ($x \leq -12$)	-0.003

Table 3: PPO + distillation hyperparameters.

Parameter	Value
LR (Stage 2/3) / (Stage 1)	3e−4 / 1e−4
PPO clip ϵ / entropy coef	0.15 / 0.0
rollout fragment / train batch	1000 / 40,000
SGD minibatch / iters	2048 / 4
workers $\times$ envs, γ/λ	8×5, 0.99/0.95
distill α decay, temp.	$0.05 \rightarrow 0$ over 8000, 1.0
teacher pool (Stage 2)	5, uniform average
total iters (Stage 3)	2000 (90M env steps)

2 Results

Table 4 reports the 10-match autograder result (the rubric metric): our agent wins all 30 graded matches with decisive goal differentials, achieving the maximum performance score of 55/55. The per-episode lift in Table 1 is concentrated at Stage 1, which raises the 031B scratch ceiling (0.880) by +0.027pp to 0.907; Stages 2–3 saturate at the same plateau with tighter variance. Ten further directions (self-play, curriculum, role-specialization, wider en-

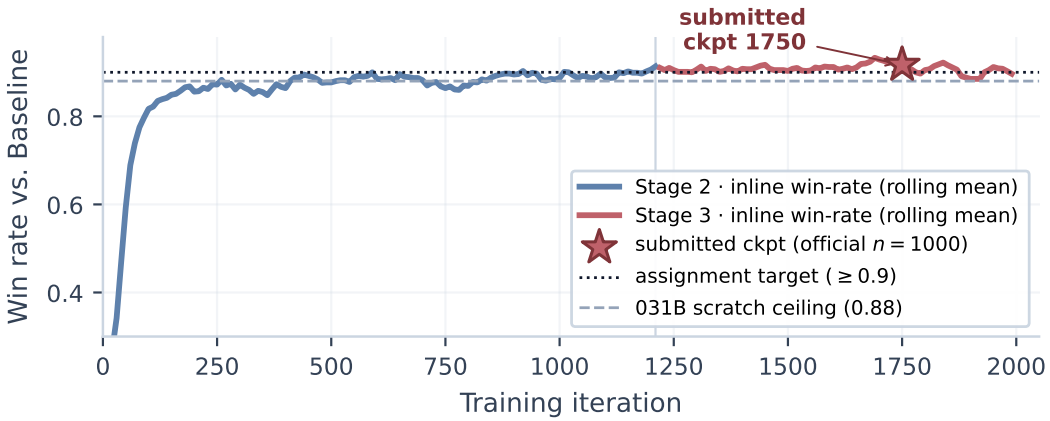

Figure 3: Per-episode win rate vs. Baseline over training iterations (dots: raw inline eval n=50/ckpt; lines: rolling mean k=5). Stage 1 distill lifts the 0.88 scratch ceiling; Stage 2–3 saturate at the $\sim$0.91 plateau. Dotted vertical: Stage 2$\rightarrow$3 handoff at iter 1210.

coders, scenario-reset warm-starts) did not beat this pipeline within statistical significance, consistent with an observation–action ceiling at $\sim$0.91.

Table 4: Autograder result (10-match protocol per opponent; total goals across all matches). Our agent clears both primary rubric thresholds and the competitive-agent bonus.

Opponent	Matches won	Goals for / against	Goal ratio	Rubric	Score
Random	10/10	431/2	215.5 : 1	$\geq$ 9/10	25/25
Baseline (`ceia`)	10/10	254/27	9.4 : 1	$\geq$ 9/10	25/25
TA competitive agent	10/10	209/10	20.9 : 1	$\geq$ 9/10 bonus	5/5
				Total performance score	**55/55**

Honest audit of the headline number. A late 25-checkpoint sweep initially put Stage 3 at 0.916, but that estimate was selection-biased: it was the max of eight within-1σ candidates with $\approx \pm 0.01$ per-sample SE. A fresh 5000-episode audit of `055v2_extend@1750` centered at 0.9066, and our own 9000-episode evaluator confirmed 0.907 ± 0.004 (CI $[0.899, 0.914]$). Stage 1 already reached 0.907 on 2000 episodes and Stage 2 0.909 on 3000; Stage 3 neither raises the mean nor the CI upper bound. The honest reading is therefore not "we found a stable 0.916 agent"; it is "we found a single, robust, autograder-clearing policy at the empirical $\sim$0.91 frontier, with Stage 3 chosen for variance (SE $\downarrow$ 2$\times$ from Stage 1), not for a higher mean."

3 Conclusion

Our two modifications are minimal: low-magnitude v2 shaping plus a Born-Again-style recursive distillation pipeline. The pipeline converts a sub-best-member deploy-time ensemble into a single-forward-pass student at 0.907, and late-extended training selects the lowest-variance checkpoint rather than a flashier higher-mean one. The $\sim$0.91 plateau is robust across $\sim$9 orthogonal paradigms we tried; further gains likely require an observation- or action-space bottleneck rather than more tuning. All training scripts, experiment snapshots, and evaluation logs are released in the project repository [8].

Acknowledgments

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